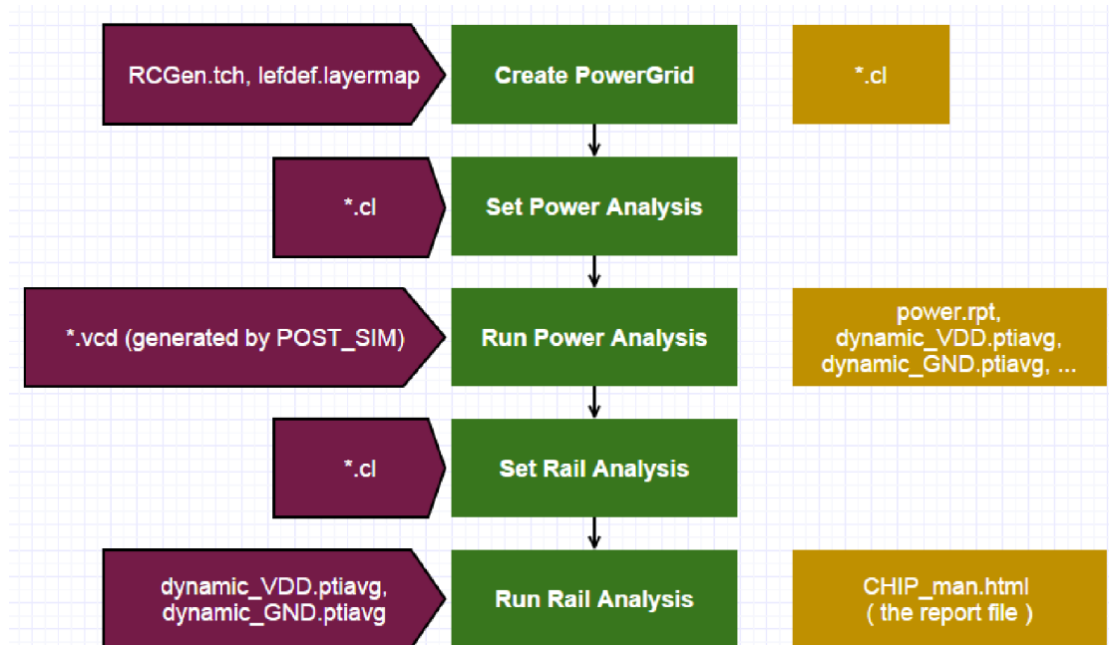


NCTU-EE IC Design LAB - 2024 Fall

Lab12 Power Rail Analysis Practice Tutorial

1. Flow overview

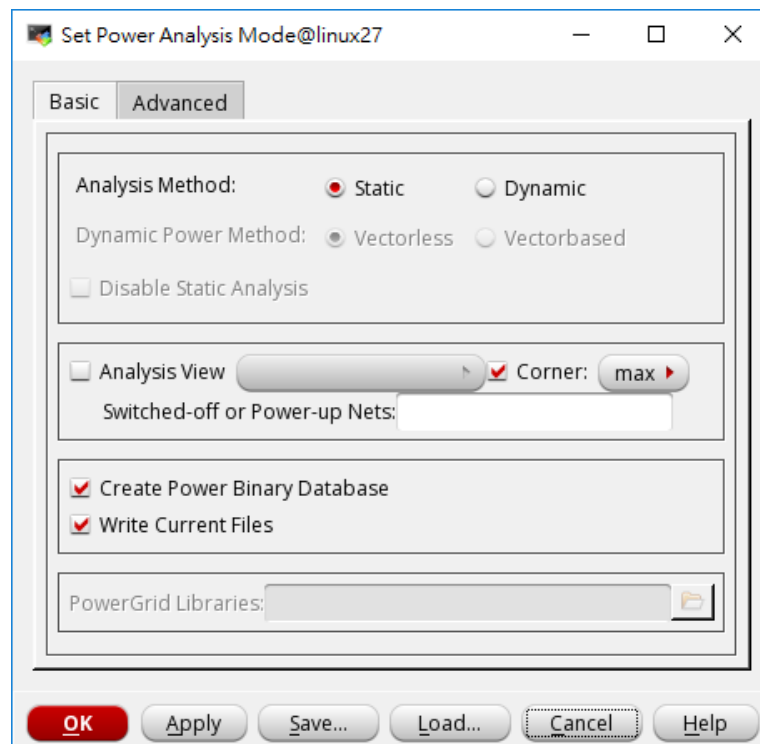
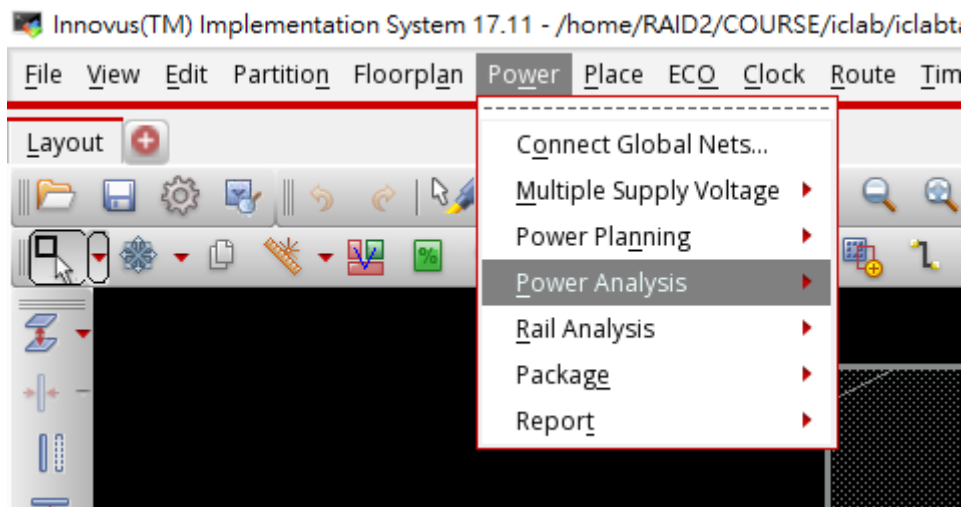


2. Set environment

- `unix% tar -xvf ~iclabTA01/Lab12.tar`
- `unix% cd Lab12/Exercise/05_APR`
- `unix% mkdir power_log` (You will save all the things here)
- `unix% innovus`
- Restore the design **DBS/CHIP.inn**

3. Static Power Analysis

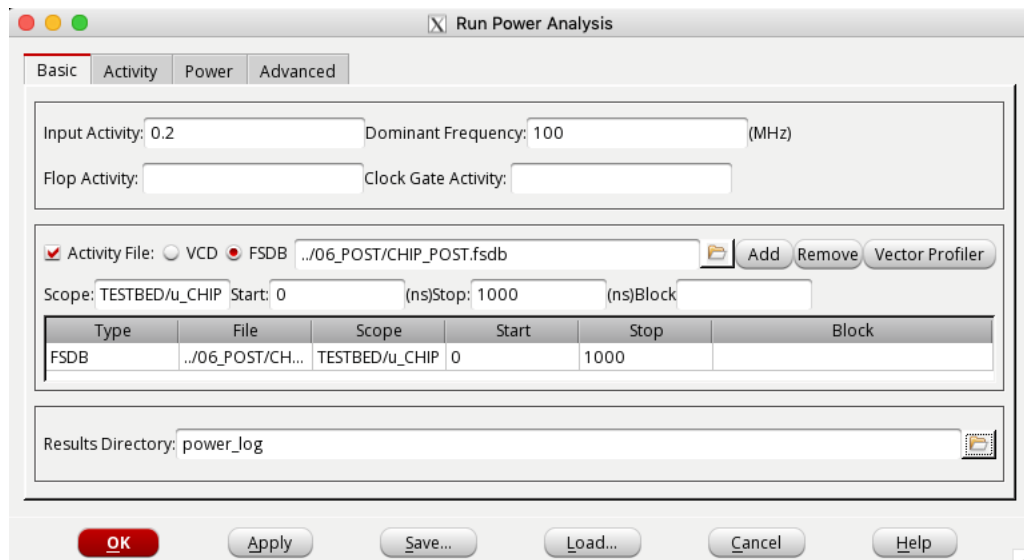
1. Save CHIP.v
2. Write CHIP.sdf
3. Run post simulation at 06_POST, the generated waveform CHIP_POST.fsdb will be used for power rail analysis.
4. In the innovus menu, open **Power** -> **Power Analysis** -> **Setup**



- i. Click **OK**.

5. In the innovus menu, open **Power -> Power Analysis -> Run**

- i. ◆Activity FILE ◆FSDB
- ii. Fill the information:
 - Select CHIP_POST.fsdb (from 06_POST)
 - Scope: TESTBED/u_CHIP
 - Start: 0; Stop: 1000
 - Press **Add**
- iii. Results Directory: power_log
- iv. Click **OK**.

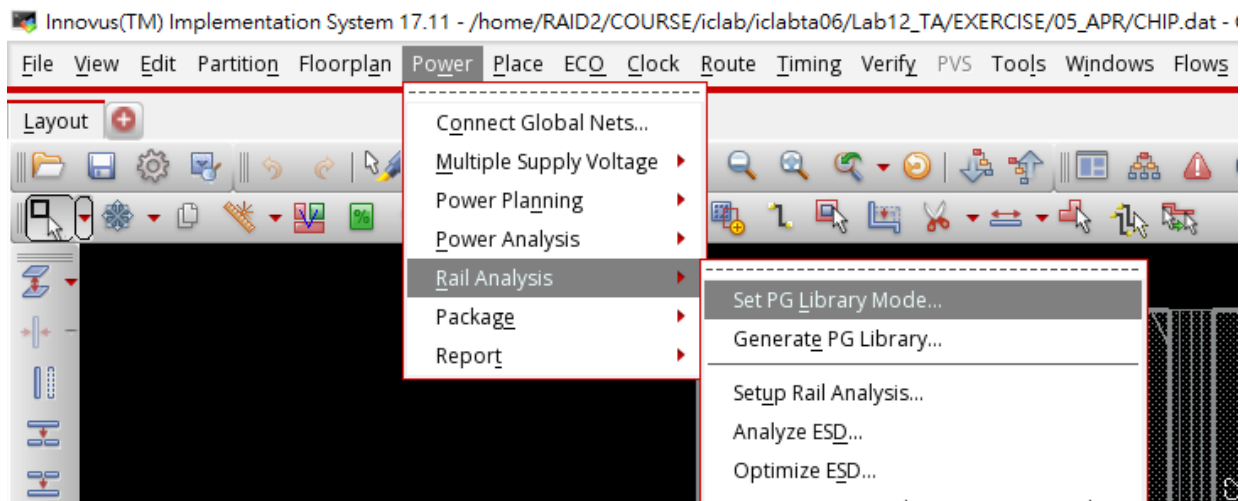


v. Results appear at terminal

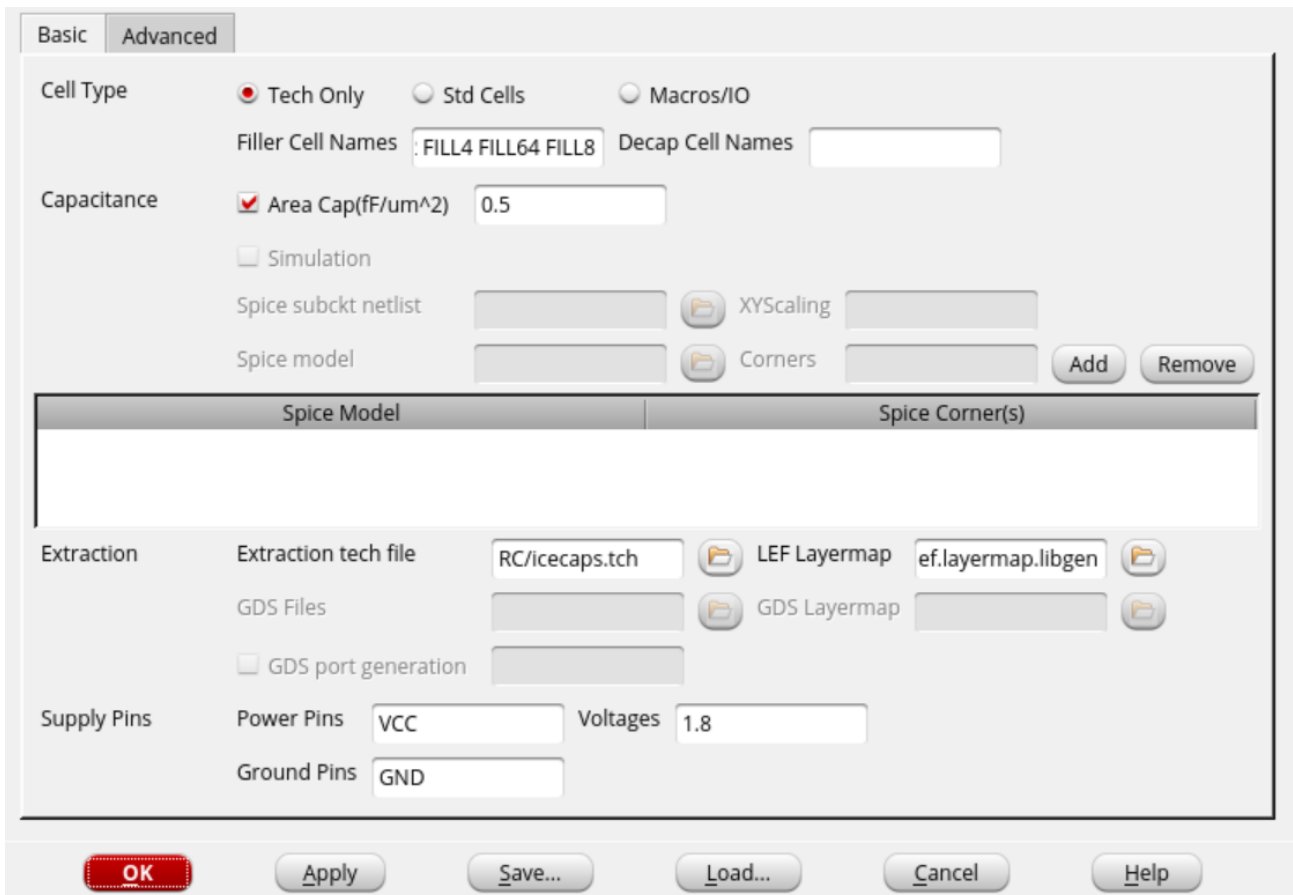
```
Total Power
-----
Total Internal Power:      56.34742919      89.3507%
Total Switching Power:     6.69368832      10.6142%
Total Leakage Power:       0.02211333       0.0351%
Total Power:               63.06323143
```

4. Create Power Grid Library

1. In the innovus menu, open **Power** -> **Rail Analysis** -> **Set PG Library Mode**

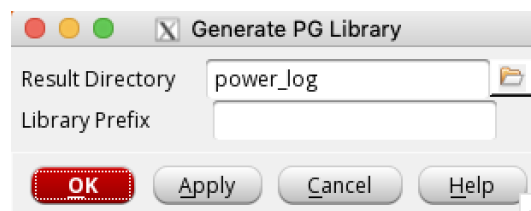


- i. Cell type: **◆Tech Only**
- ii. Filler Cell Names: **FILL1 FILL16 FILL2 FILL32 FILL4 FILL64 FILL8**
- iii. Extraction
 - Extraction tech file: **05_APR/RC/icecaps.tch** (File of type: All files(*))
 - LEF Layermap: **05_APR/layermap/lefdef.layermap.libgen**(File of type: All files(*))
- iv. Supply Pins
 - Voltages: **1.8**
 - Power pin: **VCC**
 - Ground pin: **GND**
- v. Click **OK**



2. In the innovus menu, open **Power -> Rail Analysis -> Generate PG Library**

- i. Choose power_log
- ii. Click OK



- iii. Check if the directory **techonly.cl** exists (under power_log/)

3. In the innovus menu, open **Power -> Rail Analysis -> Set PG Library Mode**

- i. Cell type: ♦Std Cells
- ii. Filler Cell Names: **FILL1 FILL16 FILL2 FILL32 FILL4 FILL64 FILL8**
- iii. Extraction
 - Extraction tech file: **05_APR/RC/icecaps.tch**
 - LEF Layermap: **05_APR /layermap/ lefdef.layermap.libgen**
- iv. Supply Pins
 - Voltage: **1.8**
 - Power pin: **VCC**
 - Ground pin: **GND**
- v. Click **OK**.

The screenshot shows the 'Set PG Library Mode' dialog box with the following settings:

- Basic** tab selected.
- Cell Type:** ☒ Std Cells
- Filler Cell Names:** FILL4 FILL64 FILL8
- Decap Cell Names:** (empty)
- Capacitance:**
 - ☒ Area Cap(fF/um^2) 0.5
 - ☐ Simulation
- Spice subckt netlist:** (empty)
- XYScaling:** (empty)
- Spice model:** (empty)
- Corners:** (empty)
- Buttons:** Add, Remove
- Table:**

Spice Model	Spice Corner(s)
-------------	-----------------
- Extraction:**
 - Extraction tech file:** RC/icecaps.tch
 - LEF Layermap:** ef.layermap.libgen
 - GDS Files:** (empty)
 - GDS Layermap:** (empty)
 - ☐ GDS port generation
- Supply Pins:**
 - Power Pins:** VCC
 - Voltages:** 1.8
 - Ground Pins:** GND
- Buttons:** OK (highlighted), Apply, Save..., Load..., Cancel, Help

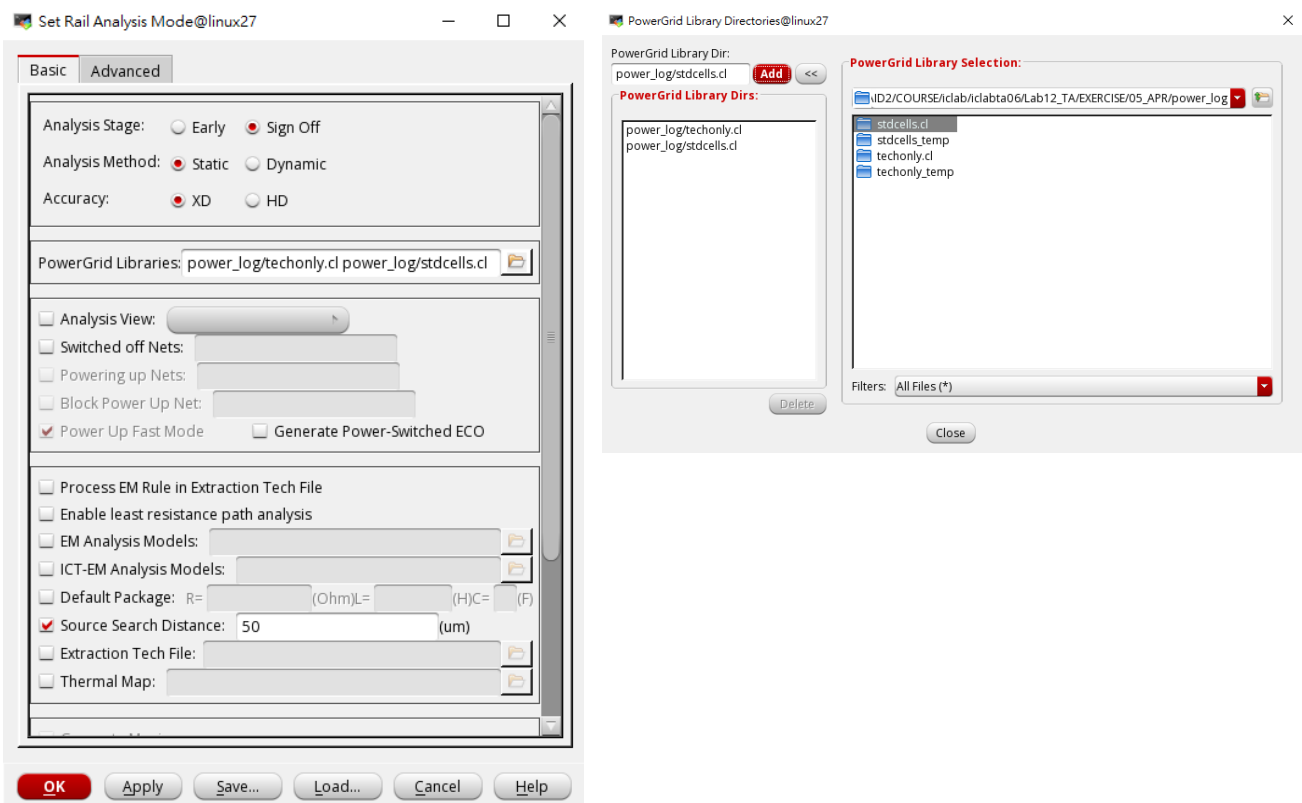
4. In the innovus menu, open **Power -> Rail Analysis -> Generate PG Library**

- i. Click
- ii. Check if the directory **stdcells.cl** exists (under power_log/)

5. Rail Analysis

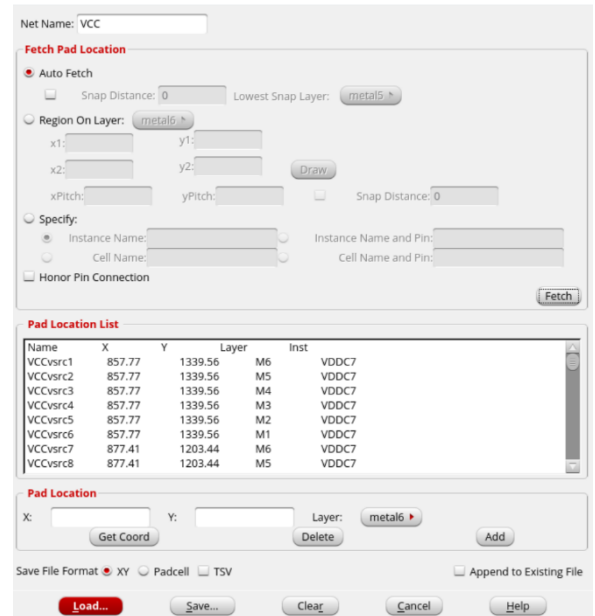
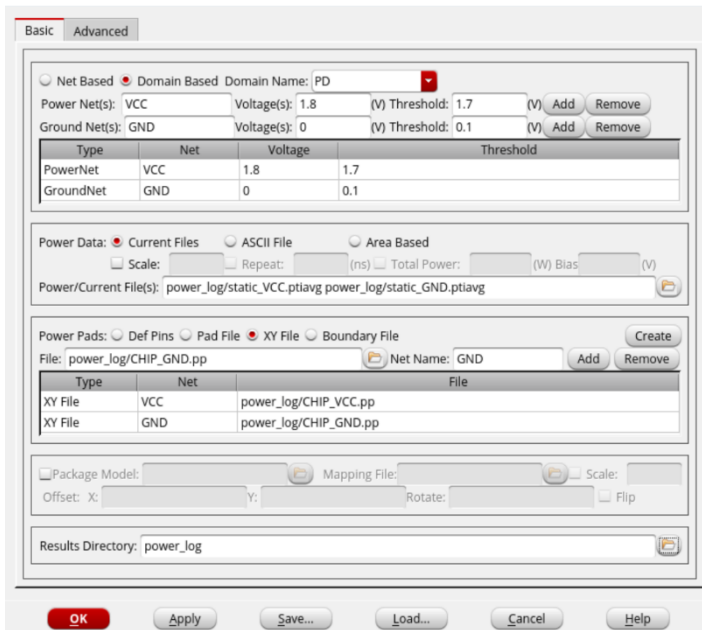
1. In the innovus menu, open **Power -> Rail Analysis -> Setup Rail Analysis**

- i. Analysis Method: **Static**
- ii. PowerGrid Libraries:
power_log/technoly.cl (should be added first)
power_log/stdcells.cl
- iii. Click



2. In the innovus menu, open **Power -> Rail Analysis -> Run Rail Analysis**

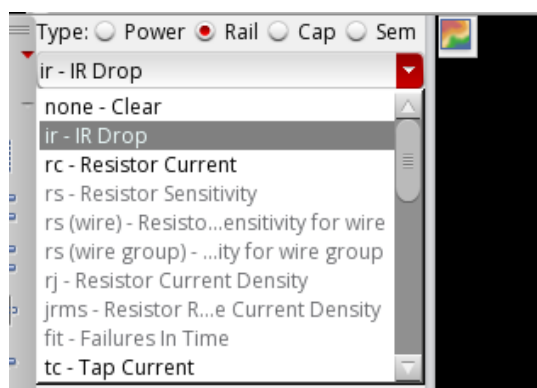
- i. ◆ Domain Based Domain Name: PD
- ii. Power Net(s): VCC Voltage(s): 1.8 Threshold:1.7 (Press **ADD**)
Power Net(s): GND Voltage(s): 0 Threshold:0.1 (Press **ADD**)
($1.8v * 5\% \approx 0.1v$)
- iii. Power/Current Files(s):
power_log/static_VCC.ptiavg
power_log/static_GND.ptiavg
- iv. Power Pads: ◆ XY File
- v. Click **Create**
 - Net Name: VCC
 - Click **Fetch**
 - Save as **power_log/CHIP_VCC.pp**
- vi. Click **Create** again
 - Net Name: GND
 - Click **Fetch**
 - Save as **power_log/CHIP_GND.pp**
- vii. Click **Cancel**
- viii. File: **power_log/CHIP_VCC.pp** Net Name: VCC (press **ADD**)
- ix. File: **power_log/CHIP_GND.pp** Net Name: GND (press **ADD**)
- x. Results Directory: **power_log**
- xi. Click **OK**

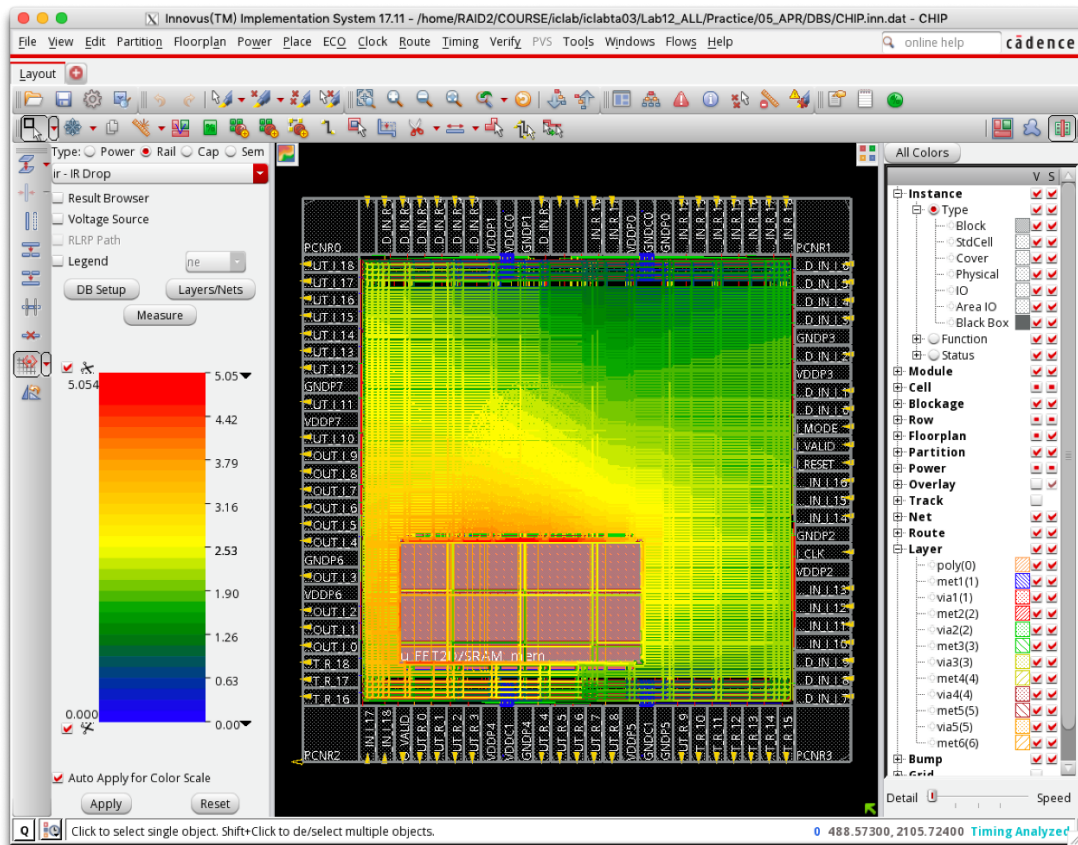


6. Power & IR Drop Results

1. In the innovus menu, open **Power -> Report -> Power & Rail Result**

- i. ◆Auto Apply for Color Scale
- ii. Click **DB Setup**
 - Power Databas: power_log/power.db
 - Rail Database: power_log/**PD_25C_avg_1**
 - Click **OK**
- iii. Type: ◆Rail
- iv. Choose **ir – IR Drop**





i. ♦Result Browser

The following shows the distribution of IR Drop (they should in the range **0.1V**)

